

Separable Differential Equations

Today's Goals

- ◆ Recognize when a first order equation $\frac{dy}{dx} = H(x, y)$ is **separable**.
- ◆ Separate variables and integrate to find general and particular solutions.
- ◆ Use initial conditions and the sign of y to resolve absolute values and constants.
- ◆ Apply separable equations to exponential growth, radioactive decay, compound interest, and Newton's Law of Cooling.

A first order differential equation $\frac{dy}{dx} = H(x, y)$ is called **separable** if we can write $H(x, y) = g(x)h(y)$. In that case we separate the variables and integrate:

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)} dy = g(x) dx \implies \int \frac{1}{h(y)} dy = \int g(x) dx.$$

Separating Variables

1 Solve the initial value problem $\frac{dy}{dx} = -8xy$, $y(0) = 4$.

Solution

Here $H(x, y) = (-8x)(y) = g(x)h(y)$, so the equation is separable:

$$\begin{aligned} \frac{dy}{y} &= -8x dx, & \int \frac{dy}{y} &= \int -8x dx, \\ \ln |y| &= -4x^2 + c_3, & |y| &= e^{-4x^2} \cdot e^{c_3} = Ae^{-4x^2}. \end{aligned}$$

This is the general solution. Applying $y(0) = 4$:

$$4 = Ae^{-4(0)^2} = Ae^0 = A, \quad \text{so} \quad |y| = 4e^{-4x^2}.$$

Since $y(0) = 4 > 0$, we have $y > 0$ near $x = 0$, so $y = |y|$ and

$$y = 4e^{-4x^2} \quad (\text{particular solution}).$$

2 Solve the initial value problem $\frac{dy}{dx} = \frac{5-4x}{y(4y^2+2)}$, $y(1) = 2$.

Solution

Writing the right side as $(5 - 4x) \frac{1}{4y^3 + 2y} = g(x)h(y)$, the equation is separable:

$$\begin{aligned}(4y^3 + 2y) dy &= (5 - 4x) dx, \\ \int (4y^3 + 2y) dy &= \int (5 - 4x) dx, \\ y^4 + y^2 &= 5x - 2x^2 + c_3 \quad (\text{general solution}).\end{aligned}$$

This cannot easily be solved for y in terms of x ; it defines a family of curves. Applying $y(1) = 2$:

$$2^4 + 2^2 = 5(1) - 2(1)^2 + c_3 \implies 20 = 3 + c_3 \implies c_3 = 17.$$

So the solution is $y^4 + y^2 = 5x - 2x^2 + 17$.

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Find the general solution to $y \frac{dy}{dx} - (8x^2 y)^{1/3} = 0$.

Solution

$$\begin{aligned}y \frac{dy}{dx} &= (8x^2 y)^{1/3} = 2x^{2/3} y^{1/3}, \\ \frac{y}{y^{1/3}} \frac{dy}{dx} &= 2x^{2/3}, \quad y^{2/3} dy = 2x^{2/3} dx, \\ \int y^{2/3} dy &= \int 2x^{2/3} dx, \\ \frac{3}{5} y^{5/3} &= \frac{6}{5} x^{5/3} + c_3.\end{aligned}$$

Multiplying by $\frac{5}{3}$ gives $y^{5/3} = 2x^{5/3} + c_4$, so

$$y = (2x^{5/3} + c_4)^{3/5}.$$

4

Find the particular solution to the initial value problem $e^y \frac{dy}{dx} = 3e^{(3x-2y)}$, $y(0) = \frac{1}{3} \ln 5$.

Solution

$$\begin{aligned}e^y \frac{dy}{dx} &= 3e^{(3x-2y)} = \frac{3e^{3x}}{e^{2y}}, \quad e^{3y} \frac{dy}{dx} = 3e^{3x}, \\ \int e^{3y} dy &= \int 3e^{3x} dx, \quad \frac{1}{3} e^{3y} = e^{3x} + c_3.\end{aligned}$$

Thus $e^{3y} = 3e^{3x} + c_4$, so $3y = \ln(3e^{3x} + c_4)$ and

$$y = \frac{1}{3} \ln(3e^{3x} + c_4) \quad (\text{general solution}).$$

Applying $y(0) = \frac{1}{3} \ln 5$:

$$\frac{1}{3} \ln 5 = \frac{1}{3} \ln(3e^0 + c_4) = \frac{1}{3} \ln(3 + c_4) \implies 5 = 3 + c_4 \implies c_4 = 2.$$

So $y = \frac{1}{3} \ln(3e^{3x} + 2)$ (particular solution).

Growth, Decay, and Cooling

Both population growth and continuously compounded interest can be modeled by the rate of change being a constant multiple of the amount: $\frac{dP}{dt} = kP$, where k is the growth rate. Separating variables gives $\frac{1}{P} dP = k dt$, and since $P(t) > 0$,

$$\ln P = kt + c_3 \implies P(t) = c_4 e^{kt} \implies P(t) = P_0 e^{kt},$$

where $P_0 = P(0)$. Radioactive decay obeys $\frac{dN}{dt} = -kN$ with $k \geq 0$, giving $N(t) = N_0 e^{-kt}$.

For continuous compounding, starting from A_0 dollars at annual rate r compounded n times per year, the amount after t years is $A_0 \left(1 + \frac{r}{n}\right)^{nt}$. Letting $n \rightarrow \infty$ gives $A(t) = A_0 e^{rt}$, which satisfies $\frac{dA}{dt} = rA$, $A(0) = A_0$. Newton's Law of Cooling/Heating states that the rate of change of an object's temperature T immersed in a medium of constant temperature A is proportional to $A - T$: $\frac{dT}{dt} = k(A - T)$.

- 5 The population of a town in 2010 was 100,000. In 2013 its population was 134,986. (a) Find the annual growth rate. (b) How long does it take for the population to double?

Solution

(a) With $P(t) = 100,000 e^{kt}$ and $P(3) = 134,986$:

$$134,986 = 100,000 e^{3k} \implies 1.34986 = e^{3k} \implies \ln(1.34986) = 3k \approx 0.3,$$

so $3k \approx 0.3$ and $k \approx 0.1$. The annual growth rate is 10%.

(b) Doubling means $200,000 = 100,000 e^{0.1t}$:

$$2 = e^{0.1t} \implies \ln 2 = 0.1t \implies t = 10 \ln 2 \approx 6.93 \text{ years.}$$

- 6 Carbon 14 has an annual decay rate $k = 0.0001216$. A piece of charcoal contains 58% as much Carbon 14 as a present-day sample of equal mass. What is the age of the sample?

Solution

Using $N(t) = N_0 e^{-kt}$ with $N(t) = 0.58N_0$:

$$0.58N_0 = N_0 e^{-0.0001216t} \implies \ln(0.58) = -0.0001216t \implies t = \frac{\ln(0.58)}{-0.0001216} \approx 4480 \text{ years old.}$$

- 7 Suppose you start with \$1000 in an account compounded continuously at annual rate r . After 3 years the amount has grown to \$1,116.28. Find r .

Solution

With $A(t) = A_0 e^{rt}$, $A_0 = 1000$, $A(3) = 1116.28$:

$$1116.28 = 1000 e^{3r} \implies 1.11628 = e^{3r} \implies \ln(1.11628) = 3r,$$

so $r = \frac{1}{3} \ln(1.11628) \approx 0.0367$, i.e. $r \approx 3.67\%$.

8

A roast, initially at 40°F , is placed in a 400°F oven at 5:00 pm. After 90 minutes the roast is at 150°F . (a) Find a formula for the temperature after t minutes. (b) When will the roast reach 160°F ?

Solution

(a) Solve $\frac{dT}{dt} = k(400 - T)$. Separating variables (with $400 - T > 0$):

$$\begin{aligned} \frac{1}{400 - T} dT &= k dt, & -\ln(400 - T) &= kt + c_3, \\ \ln(400 - T) &= -kt - c_3, & 400 - T &= c_4 e^{-kt}, \end{aligned}$$

so $T(t) = 400 - c_4 e^{-kt}$. Using $T(0) = 40$: $40 = 400 - c_4 \implies c_4 = 360$, giving $T(t) = 400 - 360e^{-kt}$. Using $T(90) = 150$:

$$150 = 400 - 360e^{-90k} \implies \frac{25}{36} = e^{-90k} \implies k = -\frac{1}{90} \ln\left(\frac{25}{36}\right) \approx 0.00405.$$

So $T(t) = 400 - 360e^{-0.00405t}$.

(b) Set $T(t) = 160$:

$$160 = 400 - 360e^{-0.00405t} \implies \frac{2}{3} = e^{-0.00405t} \implies t = -\frac{1}{0.00405} \ln\left(\frac{2}{3}\right) \approx 100 \text{ minutes.}$$

So the roast reaches 160°F at about 6:40 pm.